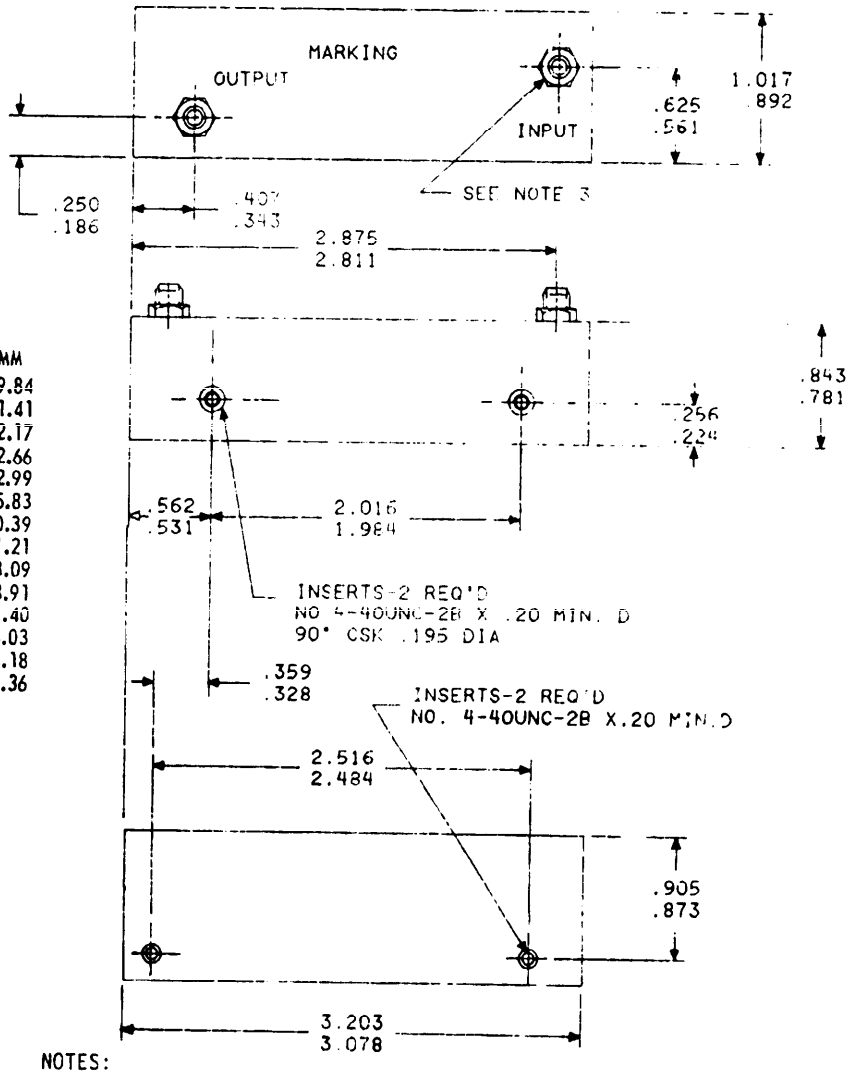
MILITARY SPECIFICATION SHEET

FILTER, BAND PASS, TYPE FR6SX22YY1

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filter described herein shall consist of this document and the latest issue of Specification MIL-F-18327.

Ⓑ THIS SPECIFICATION IS INACTIVE FOR NEW DESIGN AFTER THE DATE OF THIS SPECIFICATION



INCHES	MM	INCHES	MM
.186	4.72	.781	19.84
.224	5.69	.843	21.41
.250	6.35	.873	22.17
.256	6.50	.892	22.66
.328	8.33	.905	22.99
.343	8.71	1.017	25.83
.359	9.12	1.984	50.39
.407	10.34	2.016	51.21
.531	13.49	2.484	63.09
.561	14.25	2.516	63.91
.562	14.27	2.811	71.40
.625	15.88	2.875	73.03
		3.078	78.18
		3.203	81.36

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Miniature coaxial receptacles (female) 2 places.

FIGURE 1. Case and mounting dimensions.

Ⓑ denotes changes

FSC 5915

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Metal.

Terminals: Coaxial connector.

Weight: .25 pound.

Terminal strength: Not applicable.

Dielectric withstanding voltage: Not applicable, (circuit internally grounded to case).

Insulation resistance: Method 302, test condition A, MIL-STD-202. Not less than 1,000 megohms when measured between the output terminal center conductor and case.

Stability at temperature extremes:

1 hour at $-55^{\circ}\ ^{+0^{\circ}}_{-3^{\circ}}$ C.

1 hour at $125^{\circ}\ ^{+3^{\circ}}_{-0^{\circ}}$ C.

Electrical characteristics - As specified in table I.

Electrical characteristics: See table I.

Life: Method 108, test condition F, MIL-STD-202.

Test temperature $125^{\circ}\ ^{+10^{\circ}}_{-5^{\circ}}$ C.

Signal input level - 1 Vrms to source resistor.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure - Not applicable.

Insulation resistance - Not less than 1,000 megohms between the output terminal center conductor and case.

Electrical characteristics - As specified in table I.

Temperature rise: Not applicable.

Vibration; high frequency: Method 204, test condition B, MIL-STD-202.

Electrical load - 1 Vrms to source resistor.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure - Not applicable.

Electrical characteristics - As specified in table I.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure - Not applicable.

Electrical characteristics - As specified in table I.

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Thermal shock: Method 107, test condition A, MIL-STD-202, except that the temperature at step 3 shall be 125°C.

Exposure time - 1/2 hour.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure - Not applicable.

Insulation resistance - Not less than 1,000 megohms between the output terminal center conductor and case.

Immersion: Method 104, test condition B, MIL-STD-202. Immerse for 15 minutes in saturated solution of sodium chloride and water.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure - Not applicable.

Insulation resistance - Not less than 1,000 megohms between the output terminal center conductor and case.

Moisture resistance: Method 106, MIL-STD-202.

Polarization voltage: Not applicable.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure - Not applicable.

Insulation resistance - Not less than 1,000 megohms between the output terminal center conductor and case.

(P) Part number: M18327/44-(dash number from table I).

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TABLE I. Electrical characteristics.

Type designation	Dash number	Impedance (ohms)		Resonating capacity (pF)		Reference frequency (kHz)	Insertion loss (dB)	Frequency range (kHz)	Discrimination (dB)		DC operating voltage (volts)
		Source	Load	Source	Load				Min	Max	
FR6SX22YY1	001	5,600 ±.5%	100K ±.5%	32	49	508 approx 1/	0±1 2/	410	60		Zero
								430	60		
								450	6		
								457.5		6 3/	
								542.5		6 4/	
								550	6		
								580	60		
								590	60		

1/ Peference frequency is a point near 508 kHz where the input impedance is minimum. This point is located by observing the voltage across the filter input. For 1 Vrms into the source resistor, the filter input voltage at the reference frequency must be between .47 Vrms and .57 Vrms.

2/ Insertion loss is defined as the transfer voltage ratio (relax to 0±2dB at -55°C and +125°C).

3/ The average of actual 6dB down frequencies must be 500±2 kHz (500±3 kHz at -55°C and +125°C).

4/ The response variations over 80 percent of the 6dB bandwidth between adjacent peaks and valleys, 2dB maximum (3dB maximum at -55°C and +125°C). Between any peak and valley, 3dB maximum (4dB maximum at -55°C and +125°C).

Preparing activity:
Air Force - 85

(Project 5915-0232-44)

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